AI Supply Chain Trade Playbook

GPU demand, fabrication chokepoints, raw materials, and tradeable bottlenecks

Additional research synthesis | March 29, 2026

Core view: trade the capacity stack, not only the chip designer.

What this playbook is built to answer

1. Which bottlenecks are most likely to bind first as AI demand ramps further?
2. Which exact tickers best express foundry, HBM, packaging, and power constraints?
3. Which early warning indicators tend to surface before shortages hit earnings and price action?
4. How to separate clean expansion, tightening, deployment squeeze, and policy shock.

Demand is real

Top-four hyperscaler capex is projected around $630B-$650B in 2026; global semis toward $1T.

What breaks first

HBM, advanced packaging, and power/cooling are the main chokepoints in the current cycle.

What to trade

Core stack: TSM / ASML / MU. Rotate to packaging and power if deployment tightens.

Executive summary

This playbook treats AI as a capacity-stack trade rather than a single-stock story. The working thesis is that demand remains strong, but the highest-conviction bottlenecks have rotated away from raw silicon toward HBM memory, advanced packaging, and power/cooling infrastructure. When those chokepoints tighten, second-order suppliers can outperform the best-known GPU designer. [R1][R5][R6][R8][R9][R11][R16][R17]

- Demand still looks real. The largest hyperscalers were projected to spend roughly $630B-$650B in 2026, while global semiconductor sales were forecast to hit about $1T. [R1][R12][R17]
- The cleanest manufacturing bottlenecks are TSMC capacity, HBM supply, and advanced packaging. Broadcom explicitly flagged TSMC, laser, and PCB constraints; Samsung and Micron highlighted persistent memory tightness; ASE is still expanding packaging aggressively. [R6][R8][R9][R10][R11]
- The market often prices the front end first (NVDA / AMD / AVGO), then rotates into foundry and equipment (TSM / ASML / AMAT / LRCX / KLAC), then into memory (MU / SK hynix / Samsung), and finally into packaging and deployment (ASE / Amkor / substrate names / power and cooling).
- Raw-material shocks matter most when they hit concentrated process inputs such as photoresists, specialty gases, ABF substrates, and certain rare-earth materials. Those disruptions are usually better traded through the bottleneck owner than through the raw commodity itself. [R18][R21][R22][R23][R24][R26]

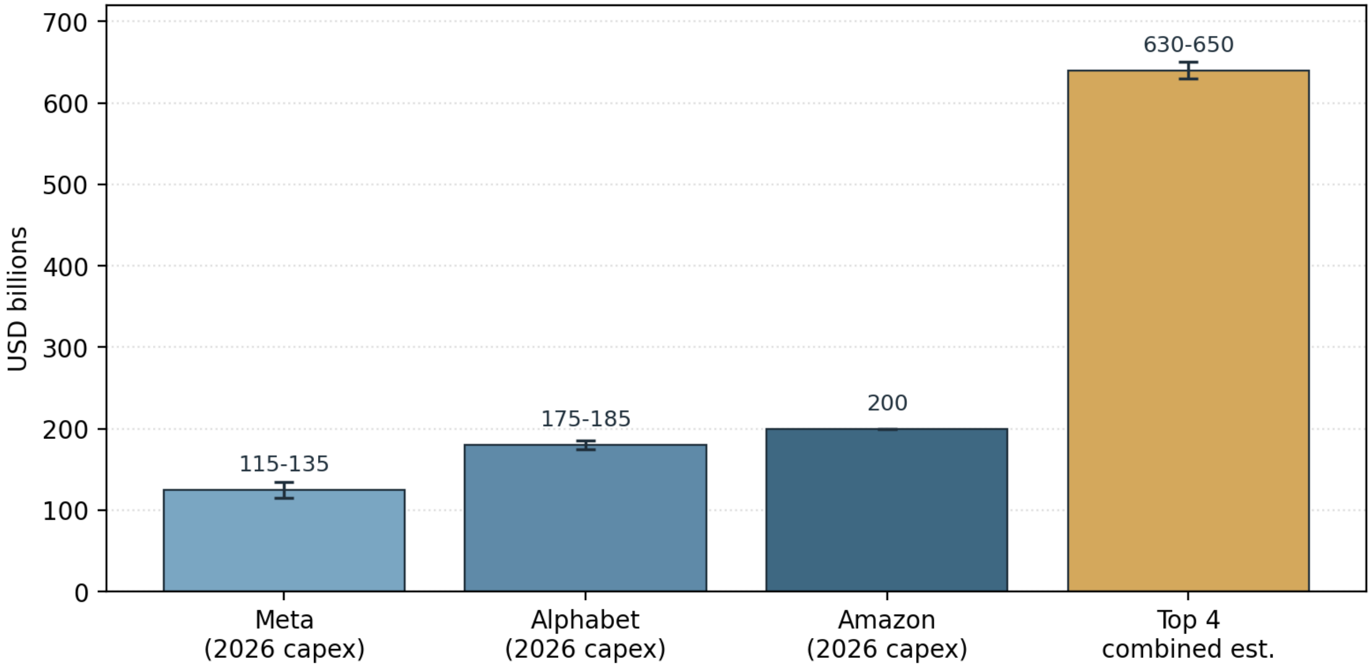
Action bias: in a clean expansion, own participation names; in a tightening phase, rotate toward memory and packaging; in a deployment squeeze, shift attention toward power, cooling, and electrification proxies. Use export-control or efficiency headlines as prompts to reassess the whole stack rather than as automatic reasons to sell everything.

Demand anchor $630B-$650B Top-four hyperscaler 2026 capex estimates. [R1][R17]	Foundry anchor $52B-$56B TSMC 2026 capex guide. [R5]
Industry anchor $1T 2026 global semiconductor sales forecast. [R12]	Tools anchor $126B / $135B Wafer-equipment sales forecast for 2026 / 2027. [R15]

Demand backdrop: the GPU cycle still has macro support

Recent reporting still points to an unusually strong AI investment cycle. The bar chart below summarizes publicly reported 2026 capex guides from Meta, Alphabet, and Amazon, plus broad estimates for the top four hyperscalers combined. The significance is not just the size of spending but its duration: TSMC is lifting capex, equipment sales are still expected to rise, and wafer demand for AI/HBM remains firm. [R2][R3][R4][R5][R12][R13][R14][R15][R17]

2026 AI infrastructure spend: disclosed plans and combined estimate

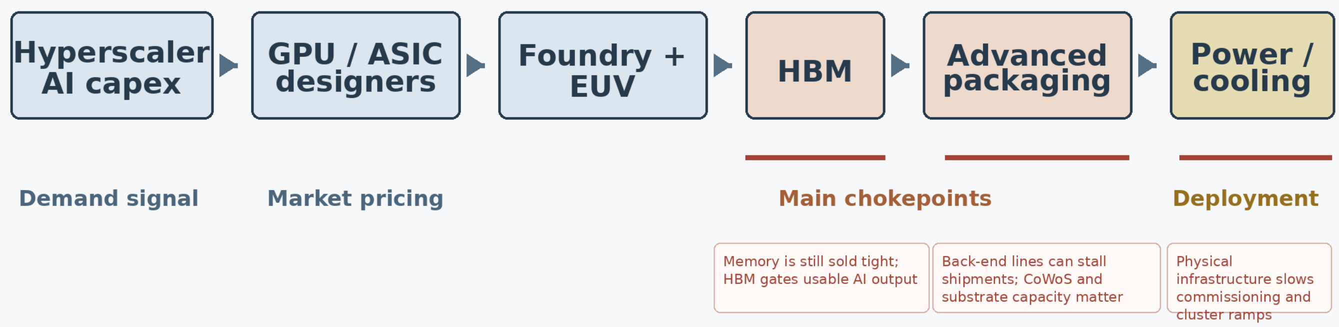


Source base: Reuters reports on Meta (Jan 28, 2026), Alphabet (Feb 4, 2026), Amazon (Feb 5, 2026), and combined top-4 estimates from Reuters / Morgan Stanley / Bridgewater (Feb-Mar 2026).

Interpretation: if hyperscaler guides keep rising while TSMC and toolmakers keep expanding, the default stance should remain constructive on the capacity stack. If capex guides flatten while memory and packaging stay tight, the trade can narrow from 'AI everywhere' to a smaller set of bottleneck owners.

How the constraint rotates through the stack

The map below captures the main analytical shift. The cycle starts with hyperscaler and platform spending, gets priced first through front-end designers, but today the highest-friction points are increasingly further downstream. Broadcom highlighted TSMC, lasers, and PCBs; Samsung and Micron highlighted memory; and Reuters reported material delays in power equipment and grid connections. [R6][R9][R10][R11][R16][R17][R18]



Analytical map from recent reporting:
Demand now collides with memory, packaging, and power more than with raw silicon alone.

- Front-end equities can move well before physical output rises. That is why NVDA / AMD / AVGO often lead on narrative and order expectations.
- Mid-cycle tightening usually shows up in TSM / ASML / HBM names first, because those are the assets that convert demand into shippable silicon. [R5][R7][R9][R11]
- When chips exist but cannot be delivered or energized, the trade migrates into packaging and then power/cooling. [R8][R16][R17]

Raw materials and process inputs: what actually matters in a disruption

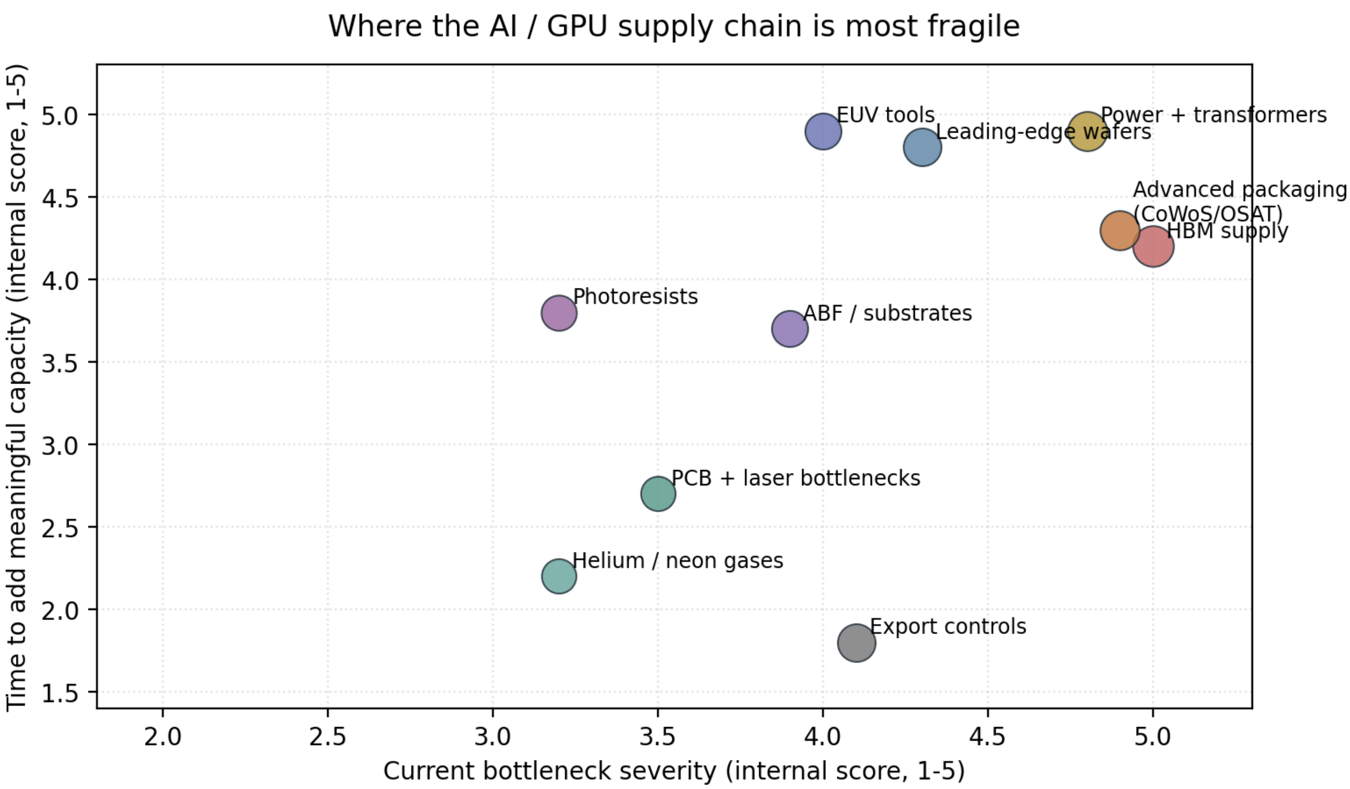
The table below answers the raw-material question directly. Some inputs are economically tiny but operationally critical. Others are abundant in principle but hard to replace at semiconductor purity levels. For traders, the better expression is usually the public company that owns the chokepoint rather than the raw material itself. [R13][R18][R21][R22][R23][R24][R26]

Input / layer	Why it matters	Current criticality	Public trade expression
Silicon wafers / epi wafers	Base substrate for advanced logic and DRAM. Demand is rising with AI logic and HBM, but raw silicon itself is not the first bottleneck today. [R13][R14][R26]	Medium	3436.T (SUMCO), 4063.T (Shin-Etsu)
Photoresists / lithography chemicals	Essential to pattern advanced nodes; supply is concentrated and technologically hard to substitute, especially at the cutting edge. [R24][R25][R26]	High	4186.T (Tokyo Ohka), 4063.T (Shin-Etsu), JSR (private)
Specialty gases: helium / neon	Low dollar share, but fabs and precision manufacturing can stop without them. Helium tightened in 2026; semiconductor-grade neon showed geopolitical vulnerability in 2022. [R18][R21]	High in disruption	LIN, APD, AIQUY, 8088.T (Iwatani)
HBM stack / DRAM process	Most practical AI-server bottleneck now. If HBM is tight, the AI GPU is less useful no matter how many logic dies exist. [R9][R10][R11][R28]	Very high	MU, 000660.KS (SK hynix), 005930.KS (Samsung)
ABF / package substrates / CoWoS	Critical to ship advanced packages. Back-end bottlenecks can strand otherwise finished chips. [R6][R8][R23]	Very high	ASX, AMKR, 3037.TW, 8046.TW, 4062.T, 6967.T, 2802.T
Power equipment / transformers / cooling	Deployment bottleneck once clusters scale. Chips can exist but revenue slips if sites cannot be energized or cooled. [R16][R17][R27]	High	VRT, ETN, NVT, HUBB
Rare-earth / niche materials	Not the main 2026 bottleneck, but shortages in yttrium and scandium show how concentrated niche inputs can surprise markets. [R22]	Low to medium	MP is an imperfect proxy; monitor rather than overweight

Bottom line: a disruption that hits photoresists, gases, HBM, substrates, or power equipment is usually more market-moving than a generic 'silicon shortage' headline. That is why the most useful watchlist often sits one or two steps away from the GPU designer.

Chokepoint severity and likely market transmission

This chart ranks chokepoints by current severity and by likely market impact if they worsen. The ranking is an analytical synthesis of recent reporting rather than a formal industry index. The key message is that the tradeable bottlenecks in 2026 are not just at the wafer stage. [R6][R7][R8][R9][R10][R11][R16][R17][R18][R19][R20]



Scores are analytical judgments synthesized from company commentary and industry reporting through Mar. 29, 2026.
Higher on both axes = a more dangerous bottleneck for the next 6-18 months.

- HBM remains the cleanest functional bottleneck: sold-tight commentary and multi-year contracts suggest scarcity is durable. [R9][R10][R11]
- Advanced packaging is the next choke point: ASE still sees its advanced-packaging business doubling in 2026, which is not what easing looks like. [R8]
- Power / transformers / cooling are becoming late-cycle bottlenecks: grid queues, long transformer lead times, and power equipment scarcity can delay revenue recognition even if chips are available. [R16][R17][R27]
- Export controls and geopolitical inputs (helium, neon, rare earths) are tail-risk accelerants rather than the base case, but they can change the tape quickly. [R18][R19][R20][R21][R22]

Exact tickers: core compute, foundry, tools, and memory

The table below is the main trading watchlist for the AI capacity stack. 'Cycle role' describes how the name tends to behave relative to the GPU cycle: Lead means it often moves on expectations, Confirm means it validates real capacity conversion, and Late means it tends to matter more once bottlenecks broaden. Ratings are qualitative, intended for positioning rather than as a substitute for security analysis.

Ticker	Company	Why it matters	Cycle role	When it tends to work best
NVDA	NVIDIA	Front-end AI demand benchmark; usually leads sentiment and order expectations.	Lead / very high beta	Clean expansion and upside revisions in AI capex.
AMD	AMD	Secondary GPU / AI compute participation; useful for breadth confirmation.	Lead / high beta	When the market broadens beyond the dominant leader.
AVGO	Broadcom	Networking / custom AI silicon exposure; management also provides good supply-chain read-through. [R6]	Lead / high beta	When hyperscaler AI networking and custom silicon demand are both firm.
TSM	TSMC ADR	Core foundry choke point; converts narrative into physical output. [R5][R6]	Confirm / very high beta	When capex rises and capacity stays tight.
ASML	ASML	EUV gatekeeper; backlog and large tool orders are strong duration signals. [R7]	Confirm / high beta	When customers are still booking future capacity aggressively.
AMAT	Applied Materials	Wafer-equipment leverage to fab buildout and process complexity. [R15]	Confirm / high beta	When capex extends through multiple quarters.
LRCX	Lam Research	Process equipment and memory exposure; benefits if DRAM/HBM spending expands. [R15]	Confirm / high beta	Memory-spending acceleration and broader fab intensity.
KLAC	KLA	Inspection / process-control leverage; quality requirements rise with complexity.	Confirm / medium-high beta	When advanced-node and HBM complexity increase.
MU	Micron	Listed U.S. HBM / memory proxy; tight supply can drive outsized earnings sensitivity. [R11]	Confirm / very high beta	HBM scarcity and pricing strength.
000660.KS	SK hynix	Best direct HBM beneficiary; record ASML order underlines long duration. [R7]	Confirm / very high beta	HBM tightening, capacity reservations, and AI server ramps.
005930.KS	Samsung Electronics	HBM plus memory scale; multi-year contracts and capex matter. [R9][R10][R28]	Confirm / high beta	When investors believe memory tightness is persistent, not cyclical noise.

Exact tickers: packaging, substrates, materials, and gases

These names matter most when the market starts asking not 'How many GPUs were ordered?' but 'How many advanced systems can actually be shipped?' Accessibility varies by brokerage because several of the cleanest exposures are local listings in Japan, Taiwan, or Korea.

Ticker	Company	Layer	What it expresses	Cycle role
ASX	ASE Technology ADR	Packaging / test	Advanced packaging capacity; ASE still expects the segment to double in 2026. [R8]	Late confirm / high beta
AMKR	Amkor	Packaging / test	Alternative advanced-packaging expression; useful when packaging becomes the narrative.	Late confirm / medium-high beta
3037.TW	Unimicron	Substrates / PCB	High-density substrate and board exposure; good when back-end shortages spread. [R6]	Late confirm / high beta
8046.TW	Nan Ya PCB	Substrates / PCB	ABF and advanced substrate read-through for AI packages. [R6]	Late confirm / high beta
4062.T	Ibiden	Package substrates	High-end package substrate proxy for advanced compute.	Late confirm / medium-high beta
6967.T	Shinko Electric	Package substrates	Another substrate-focused beneficiary if ABF / packaging stay tight.	Late confirm / medium-high beta
2802.T	Ajinomoto	ABF materials	ABF is used in nearly all high-performance computers, giving a differentiated materials angle. [R23]	Late confirm / medium beta
4186.T	Tokyo Ohka Kogyo	Photoresists	Lithography chemicals and a concentrated process input. [R24]	Process hedge / medium beta
4063.T	Shin-Etsu	Wafers + chemicals	Exposure to both silicon wafers and photoresist-related materials. [R13][R26]	Process hedge / medium beta
3436.T	SUMCO	Silicon wafers	Base-material exposure; better as a duration trade than a scarcity trade. [R13][R14]	Duration confirm / medium beta
LIN	Linde	Industrial gases	Operationally critical gas supplier; useful for disruption monitoring rather than pure AI beta.	Disruption hedge / low-medium beta
APD	Air Products	Industrial gases	Another gas-supply proxy; lower direct AI beta but relevant in a supply shock.	Disruption hedge / low-medium beta

Exact tickers: deployment bottlenecks and optional tail-risk hedges

As clusters scale, the trade can broaden into power delivery, cooling, and site-enablement names. These are later-cycle expressions: they matter most when AI demand is already proven but revenue recognition is waiting on electrical infrastructure or thermal management. [R16][R17][R27]

Ticker	Company	Why it matters	Best use in the playbook
VRT	Vertiv	Cooling, power systems, and data-center infrastructure.	Late-cycle deployment squeeze and cluster-build acceleration.
ETN	Eaton	Electrical equipment and power distribution.	Grid / transformer / switchgear bottleneck expression.
NVT	nVent	Electrical / thermal infrastructure.	Supplementary deployment basket when data-center buildout broadens.
HUBB	Hubbell	Grid and electrical components.	Useful when utility and interconnection spending becomes the story.
AIQUY	Air Liquide ADR	Industrial-gas proxy including high-purity gases.	Tail-risk hedge for gas disruptions rather than pure AI beta.
8088.T	Iwatani	Japanese industrial-gas supplier with semiconductor relevance.	Monitor around helium disruption headlines. [R18]
MP	MP Materials	Imperfect rare-earth proxy.	Tail-risk watchlist for concentrated niche-material disruptions. [R22]

Note: copper miners can capture some second-order power-buildout demand, but they are much noisier and less direct than the names above. In this playbook they sit below the preferred watchlist because the constraint is usually not ore in the ground; it is equipment, lead time, and installation.

Trading playbook: regimes, favored expressions, and invalidations

Regime	What it means	Favored expressions	What would invalidate it
1. Clean AI expansion	Demand is broad, capex guides rise, and capacity constraints are present but manageable.	NVDA / TSM / ASML as core; add AMAT / LRCX / KLAC for duration; keep MU if HBM stays tight.	Hyperscaler capex flattens, TSMC guidance cools, or AI infrastructure commentary weakens materially.
2. HBM memory squeeze	GPU demand is still strong, but the functional bottleneck is memory availability or pricing.	MU, 000660.KS, 005930.KS; keep TSM / ASML as confirmation names.	Management teams stop using sold-tight language, contracts shorten, or AI memory pricing softens sharply.
3. Packaging squeeze	Chips exist, but advanced packaging / substrates delay shipments.	ASX, AMKR, 3037.TW, 8046.TW, 4062.T, 6967.T; pair with TSM/ASML.	Back-end capacity catches up, ASE / Amkor capex peaks, or packaging lead times ease faster than expected.
4. Power and cooling squeeze	Site energization, thermal management, or transformers slow cluster deployment.	VRT, ETN, NVT, HUBB; this can outperform even if semis pause temporarily.	Grid queues shorten, transformer and cooling availability normalize, or AI buildouts are delayed for non-power reasons.
5. Export-control / geopolitical shock	Policy or conflict changes access to tools, chips, gases, or niche materials.	Favor defensive infrastructure / gas / domestic-capacity names; cut high-policy-beta exposures first.	Restrictions stabilize and no longer tighten, or affected supply is rapidly rerouted.
6. Efficiency scare / model-optimization shock	New inference efficiency headlines make investors question how much hardware intensity is truly needed.	Start by stress-testing memory and late-cycle bottleneck names; keep only the names backed by confirmed capex.	Capex guides do not change, HBM remains sold tight, and foundry/tool commentary stays strong.

Practical trading rule: do not treat every AI headline the same. Rising capex plus rising bottlenecks is bullish for the stack; rising bottlenecks plus flat capex is bullish only for a narrower set of chokepoint owners; falling capex plus easing bottlenecks is where the theme becomes vulnerable.

Early warning dashboard: what to watch before shortages hit the tape

Indicator	Why it matters	Bullish / tightening read	Warning read
Hyperscaler capex guidance	The cleanest demand signal for the whole chain. [R1][R2][R3][R4][R17]	Guides rise or stay elevated; AI remains the main capex priority.	Guides flatten or management shifts spend away from AI infrastructure.
TSMC capex and capacity language	Converts demand narrative into physical output. [R5][R6]	Capex rises, lead times remain tight, customers still nervous about supply.	Capex cools, utilization worries replace capacity worries.
HBM sold-tight language / contract length	Best direct read on functional AI-server scarcity. [R9][R10][R11]	Sold-out commentary, multi-year deals, HBM4 ramp pulled forward.	Contracts shorten, supply catches up, or pricing weakens.
ASML orders / backlog	Customers booking tools far out implies duration, not collapse. [R7][R15]	Backlog resilient, large memory-customer orders continue.	Orders defer materially or customers turn cautious on future capacity.
ASE / Amkor packaging capex	Back-end lines are what turn dies into shippable systems. [R8]	Capex rises, advanced-packaging revenue grows fast.	Packaging utilization softens or management signals faster relief.
Laser / PCB / optical lead times	Broadcom already flagged these as supply constraints. [R6]	Lead times stretch and long-term agreements become common.	Lead times compress back toward pre-tightening norms.
Helium / neon disruptions	Tiny inputs can stop production and testing. [R18][R21]	Conflict or logistics issues tighten supply suddenly.	Supply routes normalize and spot concern fades.
Transformer / grid / turbine lead times	The deployment bottleneck for data-center buildout. [R16][R17][R27]	Lead times stay extreme, power availability delays commissioning.	Utilities, transformers, and turbines become easier to source.
Export-control revisions	Policy can override fundamentals in a single headline. [R19][R20]	Restrictions tighten slowly and predictably.	Abrupt rule changes or broader country / customer scope.

Appendix A: Pine Script scanner logic

A companion TradingView script is included separately as AI_Supply_Chain_Signal_Scanner.pine. Its goal is not valuation; it is breadth and regime detection across the stack. The script scores each symbol on four conditions: (1) price above fast EMA, (2) fast EMA above slow EMA, (3) relative strength versus SOXX above its own moving average, and (4) volume above average.

- The compute bucket uses NVDA, AMD, and AVGO.
- The fab / equipment bucket uses TSM, ASML, AMAT, LRCX, and KLAC.
- The memory bucket uses MU.
- The packaging bucket uses ASE (ASX) and AMKR.
- The power bucket uses ETN and VRT.

Suggested interpretations:

- Bull regime: breadth above 70%, compute strong, and fab/equipment confirming.
- Bottleneck tightening: fab, memory, and packaging strong even if compute has stopped accelerating.
- Deployment squeeze: power names improving while fab and packaging remain firm.
- Risk-off: breadth below 40% or both compute and fab buckets deteriorate together.

The separate file is easier to paste into TradingView than a code appendix inside the PDF. For daily monitoring, use the script alongside SOXX and a manual event calendar covering TSMC, ASML, MU, NVIDIA, the hyperscalers, and major policy headlines.

Preview of the included script

```
//@version=5
indicator("AI Supply Chain Signal Scanner", overlay=false, max_labels_count=500)

// ------------------------
// USER INPUTS
// ------------------------
benchmark = input.symbol("NASDAQ:SOXX", "Benchmark ETF")

useNVDA = input.bool(true, "Include NVDA")
useAMD = input.bool(true, "Include AMD")
useAVGO = input.bool(true, "Include AVGO")
useTSM = input.bool(true, "Include TSM")
useASML = input.bool(true, "Include ASML")
useAMAT = input.bool(true, "Include AMAT")
useLRCX = input.bool(true, "Include LRCX")
useKLAC = input.bool(true, "Include KLAC")
useMU = input.bool(true, "Include MU")
useASX = input.bool(true, "Include ASE (ASX)")
useAMKR = input.bool(true, "Include AMKR")
useETN = input.bool(true, "Include ETN")
useVRT = input.bool(true, "Include VRT")
```

References

Selected sources used for the analysis below. Reuters items are especially useful because they provide management commentary, capex guides, and supply-chain bottleneck reporting that is timely enough for trading work.

[R1] Reuters, 23 Feb 2026: Bridgewater note saying the top four hyperscalers were expected to invest about $650B in 2026, up from about $410B in 2025.
[R2] Reuters, 5 Feb 2026: Amazon said 2026 capex would be about $200B.
[R3] Reuters, 4 Feb 2026: Alphabet guided 2026 capex of roughly $175B-$185B to relieve compute constraints.
[R4] Reuters, 28 Jan 2026: Meta guided 2026 capex of about $115B-$135B and still saw capacity constraints through much of 2026.
[R5] Reuters, 15 Jan 2026: TSMC guided 2026 capex of $52B-$56B and expected revenue growth near 30%.
[R6] Reuters, 24 Mar 2026: Broadcom said TSMC was hitting production limits in 2026 and called out shortages in lasers and PCBs; some optical transceiver PCB lead times had stretched from about 6 weeks to 6 months.
[R7] Reuters, 24 Mar 2026: SK hynix agreed to buy about $7.97B of ASML EUV scanners through end-2027, the largest single disclosed ASML customer order.
[R8] Reuters, 5 Feb 2026: ASE said its advanced packaging business should double to roughly $3.2B in 2026 and announced more machinery capex.
[R9] Reuters, 5 Jan 2026: Samsung described the AI-driven memory shortage as unprecedented.
[R10] Reuters, 18 Mar 2026: Samsung said it was working with major customers on 3-5 year multi-year contracts as AI chip demand stayed strong.
[R11] Reuters, 18 Dec 2025: Micron said memory markets were likely to remain tight past 2026.
[R12] Reuters, 6 Feb 2026: Global semiconductor sales were forecast to reach about $1T in 2026; advanced computing chips and memory were leading the growth.
[R13] SEMI, 10 Feb 2026: 2025 silicon wafer shipments rose 5.8%; demand was strong for advanced epitaxial and polished wafers used in AI logic and HBM.
[R14] SEMI, 28 Oct 2025: Silicon wafer shipments were projected to keep rising through 2028, with AI and HBM as key drivers.
[R15] Reuters, 16 Dec 2025: Wafer-fab equipment sales were forecast at about $126B in 2026 and $135B in 2027.
[R16] Reuters, 25 Feb 2026: AI expansion in the U.S. was being hobbled by power-infrastructure bottlenecks; the largest sites could require more than 1GW of continuous load.
[R17] Reuters Breakingviews, 26 Mar 2026: The top four hyperscalers were projected to spend about $630B in 2026, about 70% of it on servers/GPUs; gas turbines were effectively sold out until 2029 and transformer lead times in parts of Europe had reached about 100 weeks.
[R18] Reuters, 26 Mar 2026: Helium supply tightened because of Middle East conflict; Qatar supplied roughly one-third of global helium output in 2025.
[R19] Reuters, 5 Mar 2026: U.S. export-rule thresholds tightened for some AI chip installations and customer assurances.
[R20] Reuters, 11 Feb 2026: U.S. lawmakers pushed to tighten China’s access to semiconductor manufacturing tools.
[R21] Reuters, 11 Mar 2022: Ukrainian suppliers historically provided roughly 45%-54% of semiconductor-grade neon, illustrating how low-dollar gases can become mission-critical bottlenecks.
[R22] Reuters, 26 Feb 2026: Shortages in yttrium and scandium were worsening and supply remained heavily concentrated in China.
[R23] Ajinomoto Build-up Film (ABF) company materials: ABF is used in nearly all high-performance computers, making it a useful proxy for substrate dependency.
[R24] Tokyo Ohka Kogyo company materials / investor materials: semiconductor photoresists are a core lithography input and TOK reports a leading global share.
[R25] JSR company materials: JSR remains a key photoresist supplier across i-line through EUV generations, though it is no longer a public equity after its 2024 delisting.
[R26] Shin-Etsu company materials: Shin-Etsu supplies both silicon wafers and photoresist-related semiconductor materials.
[R27] Reuters, 2 Dec 2025: Power-equipment makers including Siemens, GE, Hyosung HICO and WEG were investing in transformer manufacturing capacity.
[R28] Reuters, 19 Mar 2026: Samsung planned more than 110T won (about $73B) of 2026 investment tied to its AI-chip push.